

Thesis Talk

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Overview of Topics

- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Discussion
- Outlook

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- Trying to make an accurate model for crack propagation in snow
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will probably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:

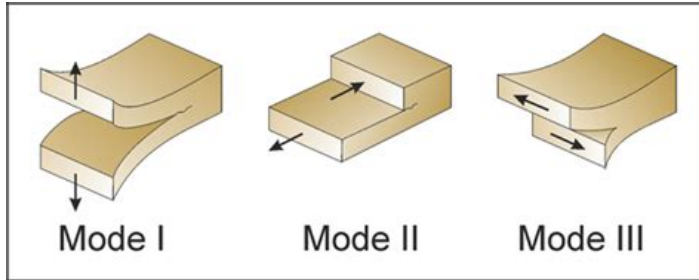


Figure: Fracture modes according to Philipp et al., 2013

Words I will probably say a lot explained II

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (Gaume et al., 2018)
- Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow (Trottet et al., 2022)
- Supershear: Cracks propagate faster than shear wave speed in snow (Trottet et al., 2022)

Overview of Models

- Model domain 400m wide, 3m high
- 1m snow layer, 2m air layer
- Multiple source versions: single source, source line with sources firing with a time delay (moving source) and rupture front

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